Quantum Mechanics II — Exercise Sheet 7

Unitary evolution and moving packets

Variant: Solutions

Exercise 1 (A group with an unbounded generator). On $\ell^2(\mathbb{N})$ let $(U_t c)_n = e^{-\mathrm{i}tn^2} c_n$. (i) Show that this is a strongly continuous unitary group. For which c does $(U_t c - c)/t$ converge in ℓ^2 as $t \to 0$? (ii) Find the generator in the convention $U_t = e^{-\mathrm{i}tH/\hbar}$.

Solution. Each phase has modulus one, and the phases multiply according to addition of times. For strong continuity split the squared norm into finitely many coordinates and a tail bounded by $4\sum_{n>N} |c_n|^2$. First choose N to make the tail small, then use continuity on the finite part. Coordinatewise differentiation forces the derivative to have entries $-\mathrm{i}n^2 c_n$. It can belong to ℓ^2 only when $\sum n^4 |c_n|^2 < \infty$. For such c, the inequality $|e^{-\mathrm{i}y} - 1| \le |y|$ gives convergence by the same finite-part and tail argument. Therefore

$$(Hc)_n = \hbar n^2 c_n, \qquad D(H) = \{c : \sum n^4 |c_n|^2 < \infty\}.$$

Exercise 2 (Locating the narrowest instant). A normalized Schwartz packet has initial variances $\Delta Q^2 = 9s^2$ and $\Delta P^2 = \hbar^2/s^2$, and symmetrized covariance $C_{QP} = -2\hbar$. Here $s > 0$ and $C_{QP} = \mathrm{Re}\langle (Q - \langle Q\rangle)\psi, (P - \langle P\rangle)\psi\rangle$. For free evolution find the time of least position variance and that variance. Check the covariance uncertainty inequality for the stated data.

Solution. The free variance rule gives

$$\Delta Q_t^2 = 9s^2 - \frac{4\hbar t}{m} + \frac{\hbar^2 t^2}{m^2 s^2} = 5s^2 + \frac{\hbar^2}{m^2 s^2}\left(t - \frac{2ms^2}{\hbar}\right)^2 .$$

The least variance is $5s^2$ at $t = 2ms^2/\hbar$. For the centered position and momentum vectors, Cauchy–Schwarz gives $\Delta Q^2 \Delta P^2 \ge |z|^2$, where $\mathrm{Re}\, z = C_{QP}$ and $\mathrm{Im}\, z = \hbar/2$. Thus the covariance uncertainty expression is

$$\Delta Q^2 \Delta P^2 - C_{QP}^2 = 5\hbar^2 \ge \frac{\hbar^2}{4}.$$

Thus the data satisfy that necessary inequality.

Exercise 3 (A packet under a constant force). Let $F \in \mathbb{R}$ and $f \in \mathcal{S}(\mathbb{R})$ be normalized. Let χ_t be its free evolution. (i) Verify by differentiation that

$$\psi_t(x) = e^{\mathrm{i}(Ftx - F^2 t^3/(6m))/\hbar} \chi_t\left(x - \frac{Ft^2}{2m}\right)$$

solves the Schrödinger equation with potential $V(x) = -Fx$ and initial vector f. (ii) Find its position and momentum means and its position variance in terms of the free packet. No assertion about a general self-adjoint domain for this Hamiltonian is required.

Solution. Put $y = x - Ft^2/(2m)$ and $S = Ftx - F^2t^3/(6m)$. The derivatives are $y_t = -Ft/m$, $S_x = Ft$ and $S_t = Fx - F^2t^2/(2m)$. After factoring out $e^{\mathrm{i}S/\hbar}$, the left side of the equation is

$$\mathrm{i}\hbar\partial_t\psi_t = e^{\mathrm{i}S/\hbar}\left[-S_t\chi_t + \mathrm{i}\hbar\partial_t\chi_t - \frac{\mathrm{i}\hbar Ft}{m}\partial_y\chi_t\right].$$

The kinetic term gives $F^2t^2\chi_t/(2m) - \mathrm{i}\hbar Ft\partial_y\chi_t/m - \hbar^2\partial_y^2\chi_t/(2m)$. Adding $-Fx\chi_t$ agrees with the displayed expression by the free equation. At $t = 0$ both the phase and the shift vanish. Changing variables and differentiating the spatial phase give

$$\langle Q\rangle_{\psi_t} = \langle Q\rangle_{\chi_t} + \frac{Ft^2}{2m}, \quad \langle P\rangle_{\psi_t} = \langle P\rangle_f + Ft, \quad \Delta Q^2_{\psi_t} = \Delta Q^2_{\chi_t}.$$

Exercise 4 (Time reversal of a free packet). For a normalized $f \in \mathcal{S}(\mathbb{R})$ let $Kf = \overline{f}$. (i) Show that $KU_t f = U_{-t}Kf$ for free evolution. Relate the means of Q and P in Kf to those in f. (ii) Explain how conjugating a packet at its narrowest instant changes its later spreading.

Solution. The Fourier transform satisfies $\widetilde{Kf}(p) = \overline{\widetilde{f}(-p)}$. Conjugating the free phase $e^{-\mathrm{i}tp^2/(2m\hbar)}$ reverses the sign of time, so $KU_t = U_{-t}K$. Position density is unchanged and momentum density is reflected. Hence

$$\langle Q\rangle_{Kf} = \langle Q\rangle_f, \qquad \langle P\rangle_{Kf} = -\langle P\rangle_f.$$

The covariance changes sign, while both initial variances remain the same. At a narrowest instant the covariance is zero. The conjugated packet then has the same position-variance curve about that instant, and the opposite mean velocity.